

A Packing of 15 Four-Spaces in $\mathbb{R}^9$

Limin Ge*

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Abstract

We give 15 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.380616. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.370046, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 15 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.380616 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 15$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.370046, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 15/14 = 2.380952$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N15.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
595154226216738763062584968005579954363306108662060878541133820911095380
519847874171446315225649139810034716933300785131563586195207705566733350
586031033632101533908967953644689028038342976752
```

denominator:

```
250000000000032454294646336673392284476918890536025297599640820951462746
845905022652414101578131792406493899901273757216307889632957184821554114
923431302835106716160125892993937589077856225197
```

B Coordinates

15 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N15.txt`.

```
-244544260103 -163450342638 241108689629 253092236620 5903863756 454181352953 122190113528 -338033307002 1195158510 -411317986688 178002994009 -563760064242
-324643219435 266266471849 85548028727 375321053261 290532786639 -501682134474 -190966153666 -109925332527 2875582 -195740604759 -109200757173 290393874509
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